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University Management System

Technical Summary

Purpose

Define the approved product direction, technology stack, core modules, engineering tools, and implementation roadmap for the University Management System.

VERSION 1.1

July 2026

Current status: Frontend prototype approved as the visual and workflow baseline.

1. Product Direction

The University Management System will be a secure, role-based ERP for administrators, faculty, and students. The approved React prototype defines the visual design, navigation, dashboards, and principal workflows. Production development will connect this experience to a .NET 8 Web API and Microsoft SQL Server.

Current prototype: responsive frontend, mock login, role dashboards, tables, forms, charts, timetable, attendance, assignments, library, results, notifications, and analytics. Mock data and authentication will be replaced in production.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Vite	Responsive component-based web application.
Navigation	React Router	Protected, role-based routes.
UI and charts	CSS, Recharts, Lucide	Design system, data visuals, and icons.
Backend	.NET 8 Web API	Business rules, validation, security, and REST APIs.
Data access	Entity Framework Core	SQL mapping, transactions, and migrations.
Database	Microsoft SQL Server	Academic and operational data storage.
Authentication	Identity / university SSO	Secure sessions, roles, and permissions.
Testing	Vitest, RTL, xUnit	Frontend, backend, and integration verification.
Deployment	IIS / Azure + CI/CD	Controlled environment-specific releases.

3. Architecture

- React communicates only with versioned .NET API endpoints.
- Controllers validate requests and delegate rules to application services.
- Entity Framework Core manages SQL Server access and migrations.
- Backend authorization is the source of truth for every protected action.
- Audit logs record important identity, attendance, marks, and data changes.

4. Core Modules

Portal	Principal scope
Administrator	Staff, students, courses, batches, subjects, timetable, examinations, library, reports, and analytics.
Faculty	Profile, subjects, timetable, students, attendance, assignments, materials, submissions, and marks.
Student	Profile, courses, timetable, attendance, assignments, materials, results, library, notices, and events.

5. Development Tools

- VS Code: React development; Visual Studio 2022: .NET development.
- Git and GitHub: version control, reviews, issues, and releases.
- Swagger and Postman: API documentation and integration testing.
- SQL Server Management Studio: database administration and diagnostics.
- GitHub Actions or Azure DevOps: automated builds, tests, and deployments.
- Serilog/Application Insights: production logging and monitoring.

6. Security and Engineering Rules

- Enforce least-privilege permissions in the backend, not only in the UI.
- Use HTTPS and secure HttpOnly cookies; never commit credentials or secrets.
- Validate all requests and use transactions for multi-record operations.
- Preserve operational history and audit sensitive academic changes.
- Every screen must support loading, empty, success, validation, and error states.

7. Delivery Roadmap

Phase	Outcome
1. Foundation	Requirements, roles, permissions, authentication, audit logs, environments, and CI/CD.
2. Masters and scheduling	Staff, student, course, batch, subject, timetable, and calendar workflows.
3. Teaching and assessment	Attendance, assignments, materials, exams, marks, results, and reports.
4. Launch	Library, notifications, security, accessibility, performance, backups, and pilot testing.

8. Immediate Next Steps

- Freeze the first-phase functional scope with university stakeholders.
- Define the SQL Server schema and role-permission matrix.
- Implement secure authentication and typed frontend API services.
- Build the first complete workflow: Student Master from UI to database.
- Pilot with selected administrators, faculty, and students.

This summary is the initial engineering baseline and should remain version controlled.